Quantum Physics

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Student Number: 12311405 **Assignment:** Number 1

Problem Derive the Sommerfeld quantization condition for a harmonic oscillator with force $F = -kq$, where q is the generalized coordinate

(a) **Lagrangian**

The Lagrangian is defined as $\mathcal{L} = \mathbf{T} - \mathbf{V}$, where **T** denotes the Kinetic energy and **V** the potential energy:

$$\mathbf{T} = \frac{1}{2}m\dot{q}^2, \quad \mathbf{V} = \frac{1}{2}kq^2$$

Thus,

$$\mathcal{L} = \mathbf{T} - \mathbf{V} = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$$

(b) **Generalized momentum**

The generalized momentum conjugate to q is

$$p = \frac{\partial \mathcal{L}}{\partial \dot{q}} = m\dot{q}$$

which gives

$$\dot{q} = \frac{p}{m}$$

.

Then, the Lagrangian can be expressed as

$$\mathcal{L} = \frac{p^2}{2m} - \frac{kq^2}{2}$$

(c) **Equation of motion**

Applying the Euler–Lagrange equation, we can give the e.o.m of the harmonic oscillator

$$\frac{d}{dt}(\frac{\partial \mathcal{L}}{\partial \dot{q}}) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

we obtain

$$m\ddot{q} + kq = 0$$

and this equation have the general solution:

$$q = A\cos(\sqrt{\frac{k}{m}}t + \phi)$$

where A is the amplitude and ϕ the initial phase, these two can be obtained from the initial condition.

(d) **Hamiltonian**

For this conservative system, the Hamiltonian equals the total energy:

$$\mathcal{H} = \sum p\dot{q} - \mathcal{L} = \frac{p^2}{2m} + \frac{kq^2}{2} = \frac{kA^2}{2}$$

Since the Lagrangian doesn't contain time t, energy is conserved Then:

$$A^2 = \frac{2E}{k}$$

(e) **Sommerfeld Quantization**

The Bohr–Sommerfeld quantization condition states $\oint p\,dq = nh$, where h is the Planck's constant and $n = 1, 2, 3, \dots$.

$$\oint p\,dq = \oint m\dot{q}\,dq = A^2k \oint \sin^2(\sqrt{\frac{k}{m}}t)\,dt = A^2\sqrt{km}\pi = nh$$

$$2n\hbar = A^2\sqrt{km} = 2E\sqrt{\frac{m}{k}}$$

$$E_n = n\hbar\sqrt{\frac{k}{m}} = n\hbar\omega$$